Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 3.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. Rearrange the following equation to give x in terms of y.

$$y = \left(\frac{x^2 + 2ax}{b}\right)^{1/2}$$

A) $x = (by^2 + a^2)^{\frac{1}{2}} - a$ B) $x = (by^2 - 2a)^{\frac{1}{2}}$ C) $x = (b^2y + a^2)^{\frac{1}{2}} - a$ D) $x = by - a$ E) $x = (by^2 + a^2 - a)^{\frac{1}{2}}$

2. Using the formula: $z = xy^2$ calculate the value of y when:

$$z = 1.2 \cdot 10^{13}$$

$$x = 3.0 \cdot 10^{-6}$$

A) $2 \cdot 10^3$ B) $4 \cdot 10^3$ C) $2 \cdot 10^9$ D) $4 \cdot 10^9$

3. The total surface area of a cylinder is numerically the same as its volume. The radius of the cylinder is r cm, the height is h cm.

Express h in terms of r.

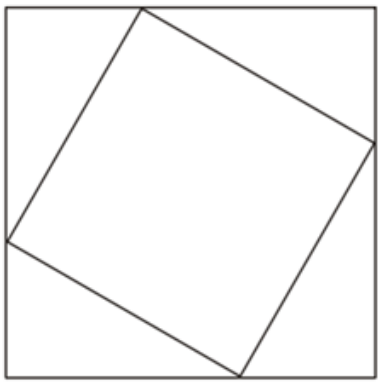
A) $h = \frac{2r}{r-2}$ B) $h = \frac{2r}{r+2}$ C) $h = r + 2$ D) $h = r - 2$ E) $h = 2r(r - 2)$

4. I have two containers with different capacities. Initially, the larger one is full of water and the smaller one is empty. I pour water from the larger container into the smaller container until they contain the same volume of water. The volume of water in the large container is now p times its capacity and the volume of water in the small container is q times its capacity.

Which one of the following statements about p and q are true?

A) $p + q = 1$ (but p and q are not necessarily both 0.5) B) $p = 0.5$ and $q = 0.5$ C) $p = 0.5$ and $q > 0.5$ D) $p > 0.5$ and $q = 0.5$ E) $p > 0.5$ and $q > 0.5$

5. A design is set up by joining the points which are one third of the way along the sides of a square. This forms a second square as shown.



This process is repeated.

Calculate the area of the fourth square as a fraction of the original square.

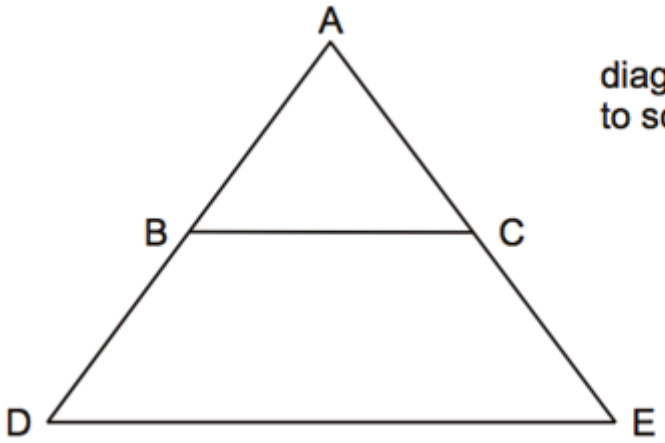
A) $\frac{1}{27}$ B) $\frac{5\sqrt{5}}{81}$ C) $\frac{125}{729}$ D) $\frac{25}{81}$ E) $\frac{5\sqrt{5}}{27}$

6. In a fairground there are two bags, each of which contains 4 coloured balls. There are a total of 4 red balls, 3 yellow balls and 1 blue ball. The player chooses one of the bags and removes two balls without replacing them. If the two balls are the same colour then the player wins. The player is equally likely to choose either bag or the balls are arranged to give the smallest possible probability for the player to win.

What is the probability that the player wins?

A) $\frac{1}{6}$ B) ¼ C) $\frac{9}{24}$ D) ½ E) $\frac{3}{4}$

7. A shape is formed by drawing a triangle ABC inside the triangle ADE. BC is parallel to DE. $AB = 4cm$, $BC = x\ cm$, $DE = x + 3\ cm$, $DB = x - 4\ cm$. Calculate the length of DE.



A) $5\ cm$ B) $7\ cm$ C) $9\ cm$ D) $4 + 2\sqrt{7}\ cm$ E) $2 + 2\sqrt{7}\ cm$

8. A solid sphere of radius r fits inside a hollow cylinder. The cylinder has the same internal diameter and the length as the diameter of the sphere. The volume of a sphere is $\frac{4}{3}\pi r^3$, where r is the radius of the sphere. What fraction of the space inside the cylinder the cylinder is taken up by the sphere?

A) $\frac{1}{4}$ B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) $\frac{2}{3}$ E) $\frac{3}{4}$

9. The design in the diagram is formed from four circles which all touches at the top of the shape (the diagram is not to scale)

The diameter of the smallest circle is d, the second is $2d$, the third is $3d$ and the largest is $4d$. Find an expression for the area of the shading in the design.



A) $\frac{5}{2}\pi d^2$ B) $4\pi d^2$ C) $¼\pi d^2$ D) $10\pi d^2$ E) $\frac{3}{2}\pi d^2$

10. The symbol $\diamond$ defines a mathematical binary operation such that $y \diamond x = \frac{y^x}{x}$ for all positive integers. What is the value of $(2 \diamond 3) \diamond 2$?

A) $2 \cdot 3^3$ B) $\frac{8}{3}$ C) $\frac{32}{9}$ D) 4 E) $2^5 \cdot \frac{\sqrt{2}}{9}$

11. The equation gives y in terms of x.

$$y = 5\left(\frac{x}{2} - 3\right)^2 - 10$$

Rearrange the equation to give an expression for x in terms of y.

A) $\pm 2\sqrt{\frac{y+10}{5}} + 6$ B) $\pm 2\sqrt{\frac{y+10}{5}} - 6$ C) $\pm 2\sqrt{\frac{y-10}{5}} + 6$

D) $\pm 2\sqrt{\frac{y-10}{5}} - 6$ E) $\pm 2\sqrt{\frac{y+50}{5}} + 6$ F) $\pm 2\sqrt{\frac{y+10}{5}} + 3$

12. A bag contains x red balls, y blue balls and z yellow balls. One ball at random is taken out and replaced. A second ball at random is taken out and replaced.

If the balls are identical in all respects except colour and are well mixed, what is the probability that the first ball was red and the second blue?

A) $\frac{(x+y)}{(x+y+z)^2}$ B) $\frac{xy}{(x+y+z)(x+z)}$ C) $\frac{xy}{(x+y+z)^2}$ D) $\frac{x+y}{x+y+z}$ E) $\frac{xy}{(y+z)(x+z)}$

13. In this question x and y are non-zero numbers. Which one of the following is **sufficient** to conclude$x < y$?

A) $x^4 < y^4$ B) $y^4 < x^4$ C) $x^{-1} < y^{-1}$ D) $y^{-1} < x^{-1}$ E) $x^{\frac{3}{5}} < y^{\frac{3}{5}}$ F) $y^{\frac{3}{5}} < x^{\frac{3}{5}}$

14. The two inequalities given below define an area on a graph.

$$y \geq x^2 + 3$$

$$x \geq \frac{1}{y}$$

Which one of the following point lies within the area defined?

A) $(-1; -6)$ B) $(2; -1)$ C) $(1; 6)$ D) $(2; 2)$

15. Evaluate:

$$\left(\sqrt{5} - \sqrt{2}\right)^2 \left(\sqrt{5} + \sqrt{2}\right)^2$$

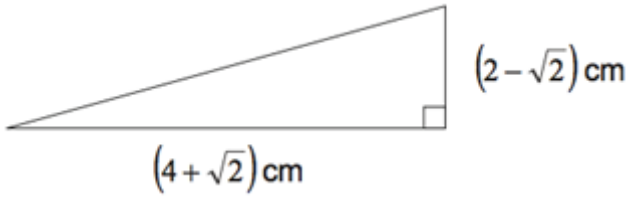
A) 9 B) $-19 - 8\sqrt{10}$ C) 49 D) $61 + 20\sqrt{10}$

16. The cross-section of a triangular prism is an equilateral triangle with side $2x$ cm. The length of the prism is d cm.

Let the total surface area of the prism be $T\ cm^2$. Given the volume of the prism is $T\ cm^3$, which one of the following is an expression for d in terms of x?

A) $\frac{x}{2x-3}$ B) $\frac{3x}{3x-2\sqrt{3}}$ C) $\frac{2x}{x-4\sqrt{3}}$ D) $\frac{2x}{x-2\sqrt{3}}$ E) $\frac{2x}{x-\sqrt{3}}$

17. The right-angled triangle shown has horizontal and vertical sides measuring $(4+\sqrt{2})$ cm and $(2-\sqrt{2})$ cm respectively.



Calculate the area of the triangle.

A) $(3-3\sqrt{2})cm^2$ B) $(3+3\sqrt{2})cm^2$ C) $(3-\sqrt{2})cm^2$
D) $(5-\sqrt{2})cm^2$ E) $(5-3\sqrt{2})cm^2$ F) $(5+3\sqrt{2})cm^2$

18. A geometric series has first term 4 and common ratio r, where $0 < r < 1$.

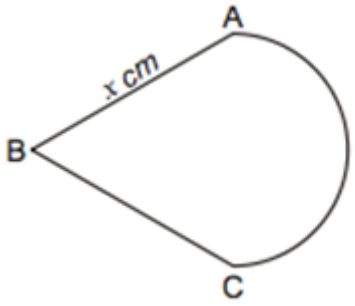
The first, second, and fourth terms of this geometric series form three successive terms of an arithmetic series.

The sum to infinity of the geometric series is

A) $\frac{1}{2}\cdot(\sqrt{5}-1)$ B) $2(3-\sqrt{5})$ C) $2(1+\sqrt{5})$ D) $2(3+\sqrt{5})$

19. In the shape shown below, A, B and C are the vertices of an equilateral triangle, side length x cm. The arc AC forms a semicircle.

Which of the following alternatives is a correct expression for the area, in cm^2, of the shape?



(not to scale)

A) $\frac{x^2(2\sqrt{3}+\pi)}{8}$ B) $\frac{x^2(\sqrt{3}+\pi)}{4}$ C) $\frac{x^2(2+\pi)}{8}$ D) $\frac{x^2(4\sqrt{3}+\pi)}{8}$

20. Simplify:

$$\left(\frac{2x^{\left(\frac{3}{2}\right)}y^3}{\sqrt{z}}\right)^2$$

A) $\frac{4x^6y^5}{z^2}$ B) $\frac{4x^3y^6}{z}$ C) $\frac{4x^3y^5}{z}$ D) $\frac{4x^{\left(\frac{5}{2}\right)}y^5}{z}$ E) $\frac{4x^{\left(\frac{5}{2}\right)}y^5}{z^2}$

21. The graphs of the following equations are drawn:

1 $y = 3x - 2$

2 $y = x^2$

3 $y = 1 - x^2$

4 $y = x + 6$

Which pair of graphs do **not** intersect?

A) 1 and 2 B) 1 and 3 C) 2 and 3 D) 2 and 4 E) 3 and 4

22. The n[th] term of a sequence is $\frac{n}{n+1}$

What is the expression for the difference between the $(n+1)^{th}$ term and the nth term?

A) $-\frac{1}{(n+2)(n+1)}$ B) $\frac{1}{(n+2)(n+1)}$ C) $\frac{n^2+n}{(n+2)(n+1)}$ D) $\frac{1}{n^2+2}$ E) $\frac{1}{n(n+1)}$

23. The sum of the roots of a quadratic equation is 7, the product of the roots is 9. What is the equation?

A) $x^2 + 7x + 9 = 0$ B) $x^2 + 7x - 9 = 0$ C) $x^2 - 7x + 9 = 0$ D) $x^2 - 7x - 9 = 0$

24. The sides of triangle ABC are as follows: $AB = 3, AC = 2, BC = 4$

Use the cosine rule, $a^2 = b^2 + c^2 - 2bc\cos A$, to find the cosine of $\angle BAC$

A) $-\frac{1}{4}$ B) $\frac{1}{16}$ C) $\frac{1}{4}$ D) $\frac{\sqrt{5}}{6}$ E) $\frac{1}{2}$

25. Solve the inequality

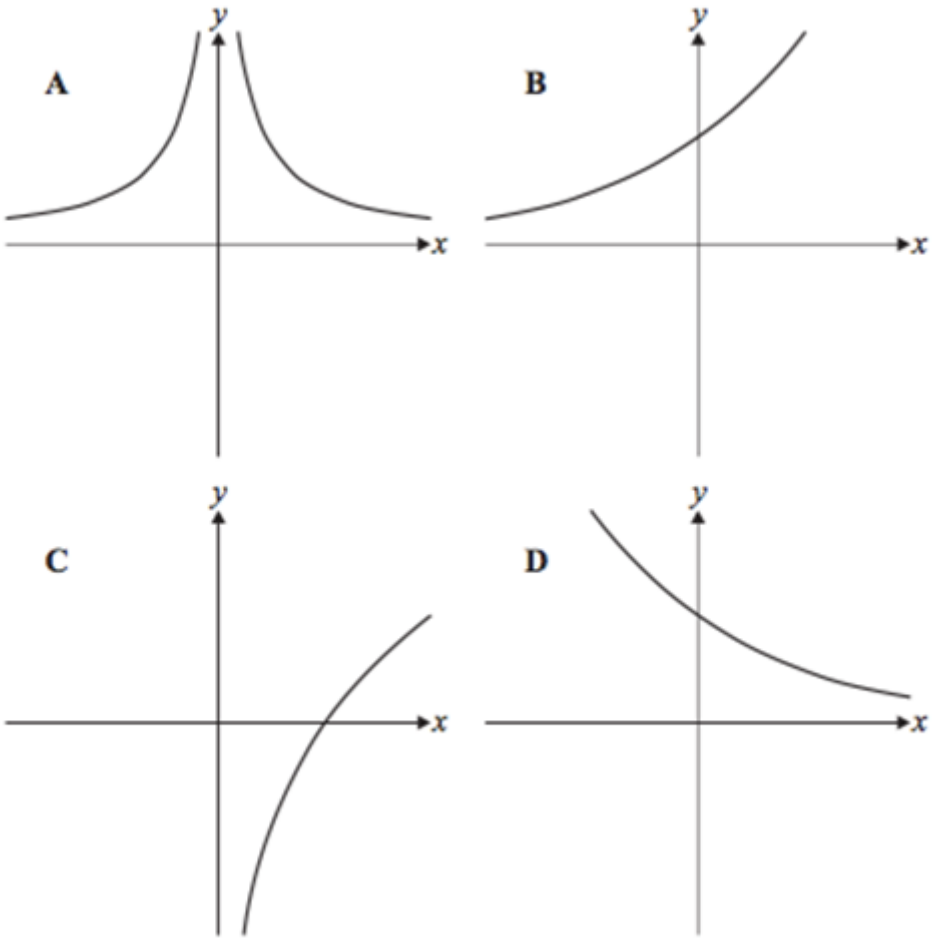
$$x^2 \geq 8 - 2x$$

A) $x \geq 4$ B) $x \leq 2$ and $x \geq -4$ C) $x \geq -2$ and $x \leq 4$ D) $x \geq 2$ or $x \leq -4$

26. $w, x, y,$ and z are integers such that $w < x^2$, $x > y^2$, $y^2 < z^2$ and $x < z$. Which one of the following **must** be true?

A) $w < x$ B) $w > y$ C) $w < z$ D) $x > y$ E) $y < z$

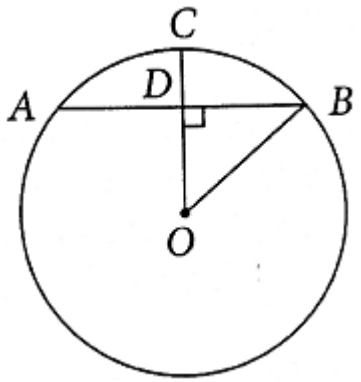
27. Which one of these graphs could be a sketch of the graph of the function $y = 2^x$?



28. Given that $x = 4.6 \cdot 10^7$ and $y = 2 \cdot 10^6$, what is the value of $\frac{x+7y}{x-2y}$?

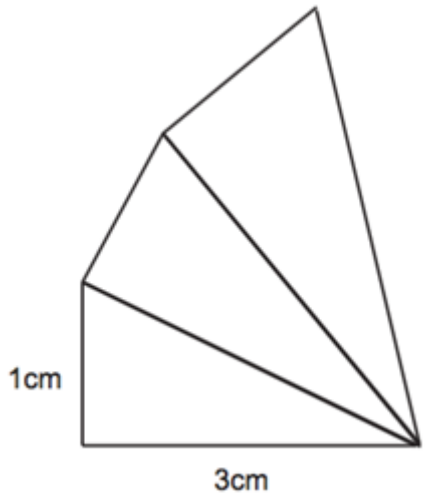
A) $\frac{10}{7}$ B) $1\frac{2}{3}$ C) 3.1 D) 7 E) 10 F) 31

29. In the figure, O is the center of the circle. If AB has a length of 16 and OB has a length of 10, what is the length of CD?



A) 2 B) 4 C) $2\sqrt{3}$ D) $8-\sqrt{35}$ E) $8-\sqrt{39}$

30. The diagram shoes three similar right-angled triangles.



What is the area of the largest triangle?

A) $\frac{5}{3}cm^2$ B) $\frac{50}{27}cm^2$ C) $5cm^2$ D) $15\ cm^2$ E) $\frac{50}{3}cm^2$

Answer key:

№	Correct Answer	Your answer
1	A	
2	C	
3	A	
4	C	
5	C	
6	A	
7	C	
8	D	
9	A	
10	C	
11	A	
12	C	
13	E	
14	C	
15	A	
16	E	
17	C	
18	D	
19	A	
20	B	
21	E	
22	B	
23	C	
24	A	
25	D	
26	E	
27	B	
28	A	
29	B	
30	A	
		Total: _____